

Augmented Reality: AR


Chutisant Kerdvibulvech, Ph.D.
Assistant Professor

ผศ.ดร.ชุตินันต์ เกดวิบูลย์เวช

Overview

-  Background
-  Definition of Augmented Reality
-  AR Applications and Computer Vision
-  AR Tools

Asst. Prof. Dr. Chutisant Kerdvibulvech

2

Augmented Reality: AR



What is Augmented Reality (AR)?



✧ A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.

✧ Generates a composite view for the user.

Asst. Prof. Dr. Chutisant Kerdvibulvech

4

What is the Goal of AR?



❧ To enhance a person's performance and perception of the world

❧ But, what is the ultimate goal?



"Augmented reality in urban places: contested content and the duplicity of code." Transactions of the Institute of British Geographers, 2012

The Ultimate Goal of AR



❧ Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce *Geek.com*

Asst. Prof. Dr. Chutisant Kerdvibulvech

6

Augmented Reality vs. Virtual Reality



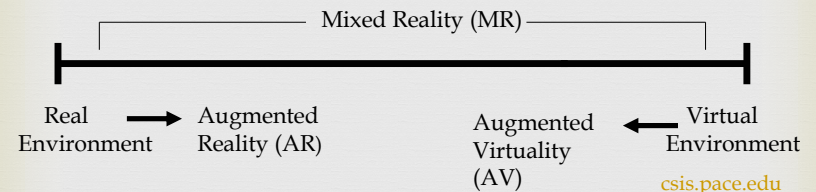
Augmented Reality

- ❧ System augments the real world scene
- ❧ User maintains a sense of presence in real world
- ❧ Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- ❧ Totally immersive environment
- ❧ Visual senses are under control of system

Miligram's Reality-Virtuality Continuum



Miligram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer

Combining the Real and Virtual Worlds



We need:

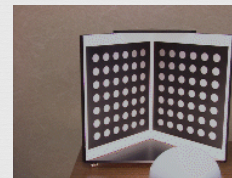
- ❧ Precise models
- ❧ Locations and optical properties of the viewer (or camera) and the display
- ❧ Calibration of all devices
- ❧ To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

Asst. Prof. Dr. Chutisant Kerdvibulvech

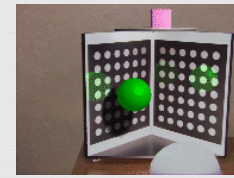
9

Combining the Real and Virtual Worlds (cont)

- ❧ Register models of all 3D objects of interest with their counterparts in the scene
- ❧ Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-reflections and virtual shading

10

Mixed Reality

Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing

1



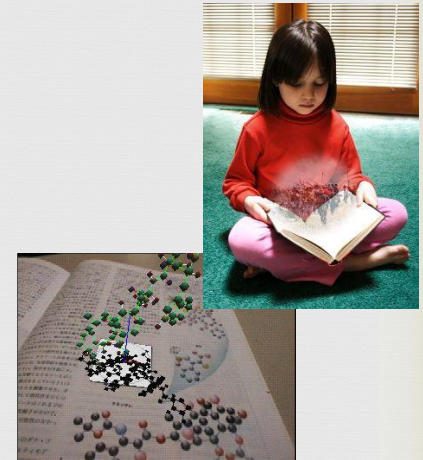
Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

11

Asst. Prof. Dr. Chutisant Kerdvibulvech

This image depicts and artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sonycsi.co.jp

12

Asst. Prof. Dr. Chutisant Kerdvibulvech

Realistic Merging



Requires:

- ❧ Objects to behave in physically plausible manners when manipulated
- ❧ Occlusion
- ❧ Collision detection
- ❧ Shadows

**All of this requires a very detailed description of the physical scene

Research Activities



- ❧ Develop methods to register the two distinct sets of images and keep them registered in real-time
 - ❧ New work in this area has started to use Computer Vision (CV) techniques
- ❧ Develop new display technologies for merging the two images

Performance Issues



Augmented Reality systems are expected:

- ❧ To run in real-time so that the user can move around freely in the environment
- ❧ Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- ❧ Update rate for generating the augmenting image
- ❧ Accuracy of the registration of the real and virtual image

Failures in Registration



Failures in registration due to:

- ❧ Noise
 - ❧ Position and pose of camera with respect to the real scene
 - ❧ Fluctuations of values while the system is running
- ❧ Time delays
 - ❧ In calculating the camera position
 - ❧ In calculating the correct alignment of the graphics camera

Applications



- ❧ Medical
- ❧ Military Training
- ❧ Hazard Detection
- ❧ Engineering Design
- ❧ Robotics and Telerobotics
- ❧ Training
- ❧ Sports
- ❧ Tourism
- ❧ Video conferencing

Asst. Prof. Dr. Chutisant Kerdvibulvech

17

Applications (cont.)



- ❧ Entertainment, Computer Games
- ❧ Manufacturing, Maintenance, and Repair
- ❧ Consumer Design
- ❧ Audio
- ❧ Education
- ❧ Architecture
- ❧ Human welfare

Asst. Prof. Dr. Chutisant Kerdvibulvech

18

Recent Augmented Reality Applications



Sixth Sense



Sixth Sense

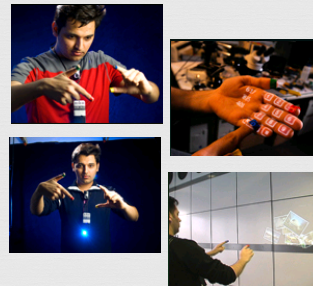


- ❧ A wearable gestural interface that augments the physical world around us with digital information
- ❧ Let us use natural hand gestures to interact with that information
- ❧ The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Asst. Prof. Dr. Chutisant Kerdvibulvech

21

Sixth Sense in 21st Century



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED



TED
TALKS

Asst. Prof. Dr. Chutisant Kerdvibulvech

22

Detail of Sixth Sense



- ☞ Sixth Sense allows people to use the internet without a screen or a keyboard
- ☞ It can turn any surface into a touch screen
- ☞ It is made with store-bought components
- ☞ *"The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques" ..from MIT Media Lab*

Asst. Prof. Dr. Chutisant Kerdvibulvech

23

Detail of Sixth Sense



- ☞ The device recognizes items in stores, and then gathers and projects information about products to the user.
- ☞ It also is able to provide quick signals to let users know what the best products are based on their needs."
- ☞ Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- ☞ Say goodbye to your encyclopedia

Asst. Prof. Dr. Chutisant Kerdvibulvech

24

Cost of Sixth Sense



- ✧ Only \$350 dollars, around 12,000 baht
- ✧ From an educator's stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- ✧ For educator's, Sixth Sense would be a way to put technology (literally) into each student's hands, because of its size and its cost.

Augmented Reality Guitar



Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel

3



Asst. Prof. Dr. Chutisant Kerdvibulvech

27

MR for Military Application



Mixed Reality for Military Application



Dr. Adrian Cheok,
National University of Singapore (NUS)

Broadcast in CNN



4



29

Augmented Reality Magic

Augmented Reality Magic

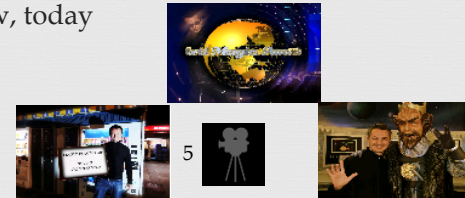
- Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- Along comes a pasteboard prestidigitator who folds augmented reality into the magic world



Asst. Prof. Dr. Chutisant Kerdvibulvech

Augmented Reality Magic (card trick)

- Blend of technology and illusion creates the magic of tomorrow, today

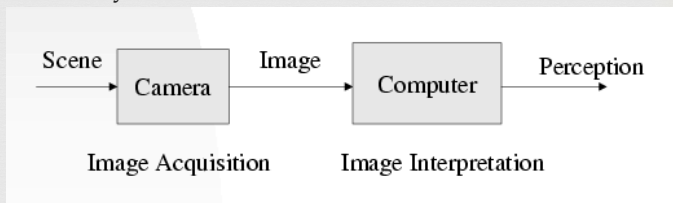


Marco Tempest,
Newmagic Communications Inc.,
United States

Computer Vision



- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



Asst. Prof. Dr. Chutisant Kerdvibulvech

33

Introduction of Computer Vision

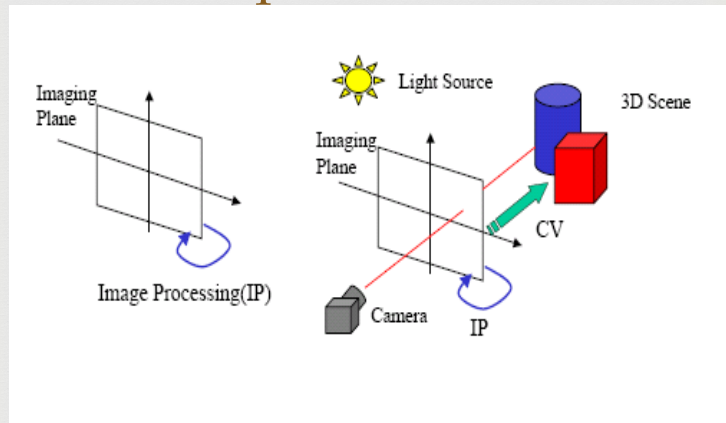


- Computer Vision (CV)
 - Emerged from AI (Artificial Intelligence) Research
 - Computational Vision proposed by Marr
 - Shape from X(X:shading, texture, contour, motion...etc.)
 - 3D Scene Understanding from 2D images
 - Robot Vision: Vision functionality for Robots
 - Media Processing, Human Interface

Asst. Prof. Dr. Chutisant Kerdvibulvech

34

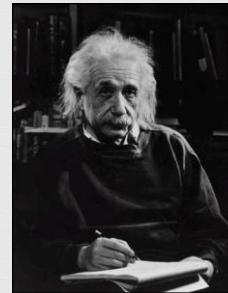
Image Processing / Computer Vision



Asst. Prof. Dr. Chutisant Kerdvibulvech

35

Computer Vision



What we see

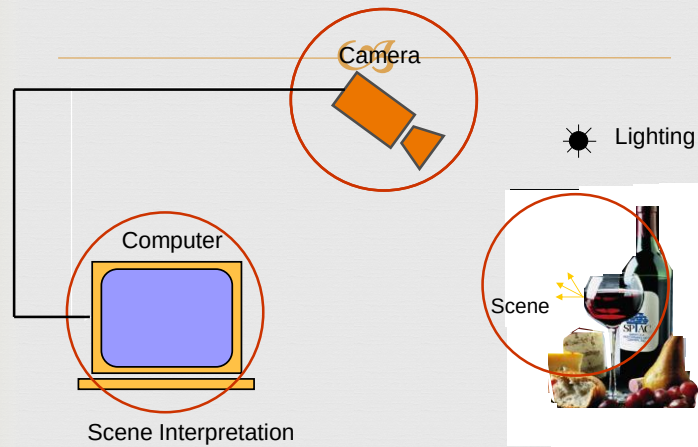
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

36

Asst. Prof. Dr. Chutisant Kerdvibulvech

Components of a Computer Vision System



Asst. Prof. Dr. Chutisant Kerdvibulvech

37

GSCM Digital Campaign (AR)



Pixlive Maker

Pixlive Player

PIXLIVE
MAKER

<http://www.vidmoti.com/>

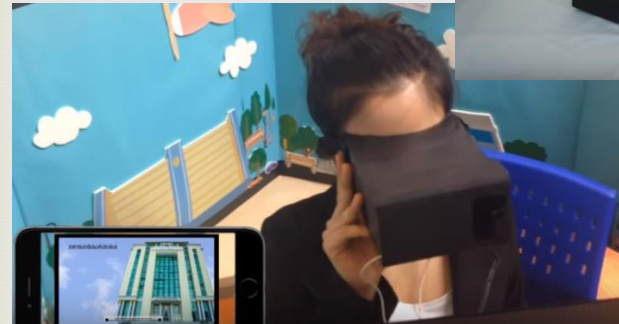


Next Poster



Chutisant Kerdvibulvech

NIDA VirtualBox



Chutisant Kerdvibulvech



Thank you

Asst. Prof. Dr. Chutisant Kerdvibulvech
Graduate School of Communication Arts and Management
Innovation,
National Institute of Development Administration (NIDA)
chutisant.ker@nida.ac.th
<https://www.facebook.com/chutisant.kerdvibulvech>